WMDR2core

Package in package 'Model'

WMD
Version 0.1 Phase 1.0 Pr
jkl created on 2026-03-06. Last modified 2026

WMDR2core diagram

Class diagram in package 'WMDR2core'

WMDR2core
Notes
1. temporalExtent shall be realized as time:interval
2. temporal<Name> elements other than temporalExtent shall be realized as an array of two arrays, one for times, the values of <Name>, e.g., temporalTerritory = [[2000-01-01, 2024-07-13], ["https://codes.wmo.int/wmdr/territory/<ISO1>", "https://codes.wmo.int/wmdr/territory/<ISO2>"]].

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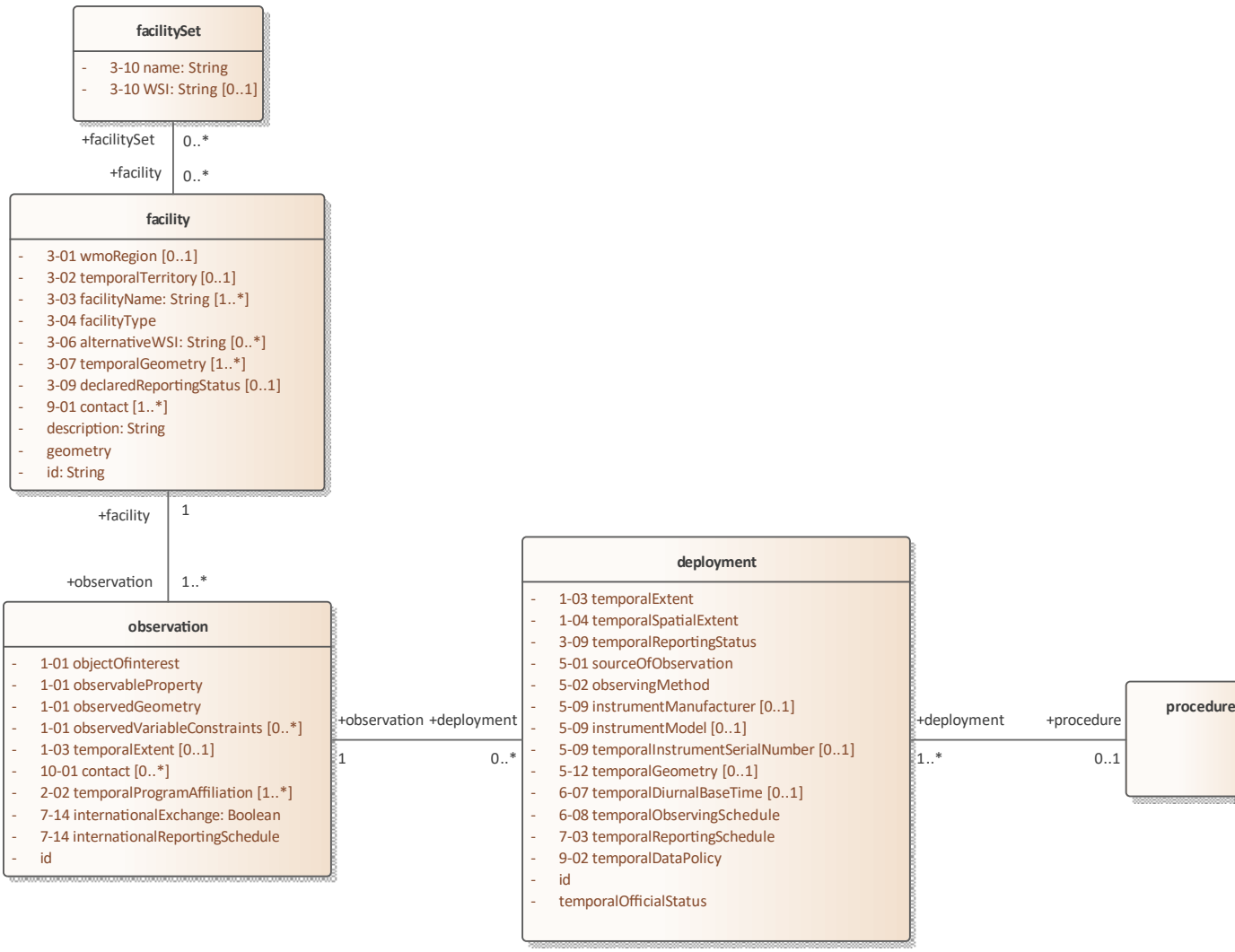

Figure 1: WMDR2core

Class in package 'WMDR2core'

depl
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ATTRIBUTES

1-03 temporalExtent : Private
Alias: records:interval

[Is static True. Containment is Not Specifi

1-04 temporalSpatialExtent : Private
Alias: records:geometry

Typical spatial geo-referenced volume covered by the observations.

EXAMPLES:
(a) Air temperature at a surface observing site: Sydney Airport, New South Wales, Australia: lat. 33.9465°N, lon. 151.1731 alt. 6.0 m above mean sea level (amsl);
(b) The projected area or volume of the cone around a particular weather radar with a maximum range of 370 km (radar reflectivity) and 150 km (Doppler); to be expressed as a geometric shape;
(c) Three-dimensional grid of radar pixels;
(d) For infrared and visible imagery by meteorological satellite (sun-synchronous): global;
(e) For nadir sounding: atmospheric column above ocean;
(f) The Japan Meteorological Agency Advanced Himawari Imager has a 10 min cycle full disk scan (0.5-2.0 km spatial resolution) located above 140.68°E geostationary orbit at an altitude of 35,786 km;
(g) River discharge by gauge: size and geometric shape of a river catchment.

Q: what is the spatial extent for a)?

[Is static True. Containment is Not Specifi

3-09 temporalReportingStatus : Private

[Is static True. Containment is Not Specifi

5-01 sourceOfObservation : Private

[Is static True. Containment is Not Specifi

5-02 observingMethod : Private

[Is static True. Containment is Not Specifi

5-09 instrumentManufacturer : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specifi

5-09 instrumentModel : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specifi

ATTRIBUTES

5-12 temporalGeometry : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

- Geospatial location of instrument/sensor
- Consider to include z-coordinate (5-05 heightAboveLocalReferenceSurface)

[Is static True. Containment is Not Specifi

6-07 temporalDiurnalBaseTime : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

indicates the end (hour) of a 24-hour period. Mandatory for 24-hour periodic observations, e.g., cumulative amount of precipitation

[Is static True. Containment is Not Specifi

6-08 temporalObservingSchedule : Private

complex type: when are observations made? Should be expressed as a temporal coverage, e.g. "from Feb-Oct on working d from 06-18 UTC"

[Is static True. Containment is Not Specifi

7-03 temporalReportingSchedule : Private

complex type: when are observations made? Should be expressed as a temporal coverage

[Is static True. Containment is Not Specifi

9-02 temporalDataPolicy : Private

remove from WMDR2 and leave for WCMP2?

[Is static True. Containment is Not Specifi

id : Private

[Is static True. Containment is Not Specifi

temporalOfficialStatus : Private

[Is static True. Containment is Not Specifi

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public deployment (Class) deployment
Cardinality: [1..*]

Target: Public procedure (Class) procedure
Cardinality: [0..1]

ObservingProcedure(s)

Association (direction: Unspecified)

facility

Class in package 'WMDR2core'

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ATTRIBUTES

3-01 wmoRegion : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

mandatory for fixed facilities, optional for mobile facilities.

[Is static True. Containment is Not Specifi

3-02 temporalTerritory : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

mandatory for fixed facilities, optional for mobile facilities

[Is static True. Containment is Not Specifi

3-03 facilityName : String Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specifi

3-04 facilityType : Private

[Is static True. Containment is Not Specifi

3-06 alternativeWSI : String Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specifi

3-07 temporalGeometry : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

must allow history

[Is static True. Containment is Not Specifi

3-09 declaredReportingStatus : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Condition: must be provided if no deployment is specified; otherwise: Optional, inferred as "best case" from deployments: temporalReportingStatus + temporalProgramAffiliation.
Exception: "closed" over-rules deployment, in which case all deployments should be terminated.

[Is static True. Containment is Not Specifi

9-01 contact : Private

ATTRIBUTES

description : String Private

[Is static True. Containment is Not Specifi

geometry : Private
Alias: 3-07 current geolocation

[Is static True. Containment is Not Specifi

id : String Private
Alias: 3-06 primaryWSI

[Is static True. Containment is Not Specifi

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public facility (Class) facility
Cardinality: [0..*]

Target: Public facilitySet (Class) facilitySet
Cardinality: [0..*]

Association (direction: Unspecified)

Source: Public facility (Class) facility
Cardinality: [1]

Target: Public observation (Class) observation
Cardinality: [1..*]

facilitySet

Class in package 'WMDR2core'

fac
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ATTRIBUTES

3-10 name : String Private

[Is static True. Containment is Not Specifi

3-10 WSI : String Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specifi

ASSOCIATIONS

observation

Class in package 'WMDR2core'

obse
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ATTRIBUTES

1-01 objectOfinterest : Private

observedVariable should be modeled using the I-ADOPT ontology approach. Combination of
- observableProperty: a state variable (e.g., temperature), or another property (e.g., concentration of something)
- objectOfInterest: e.g., dry air (of the atmosphere)
- observedVariableConstraint(s): e.g., a size fraction, a specific wavelength, a pressure (range), a temperature (range), a humidity (range), etc., as relevant for the result of the observation. Constraints shall be documented as name-value pairs, where the names shall come from a code list, and the values shall come from a code list depending on the name, or be free text.

[Is static True. Containment is Not Specifi

1-01 observableProperty : Private

observedVariable should be modeled using the I-ADOPT ontology approach. Combination of
- observableProperty: a state variable (e.g., temperature), or another property (e.g., concentration of something)
- objectOfInterest: e.g., dry air (of the atmosphere)
- observedVariableConstraint(s): e.g., a size fraction, a specific wavelength, a pressure (range), a temperature (range), a humidity (range), etc., as relevant for the result of the observation. Constraints shall be documented as name-value pairs, where the names shall come from a code list, and the values shall come from a code list depending on the name, or be free text.

[Is static True. Containment is Not Specifi

1-01 observedGeometry : Private

[Is static True. Containment is Not Specifi

1-01 observedVariableConstraints : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

observedVariable should be modeled using the I-ADOPT ontology approach. Combination of
- observableProperty: a state variable (e.g., temperature), or another property (e.g., concentration of something)
- objectOfInterest: e.g., dry air (of the atmosphere)
- observedVariableConstraint(s): e.g., a size fraction, a specific wavelength, a pressure (range), a temperature (range), a humidity (range), etc., as relevant for the result of the observation. Constraints shall be documented as name-value pairs, where the names shall come from a code list, and the values shall come from a code list depending on the name, or be free text.

[Is static True. Containment is Not Specifi

1-03 temporalExtent : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

aggregate derived from deployments if deployments are provided; otherwise, specify begin-end

ATTRIBUTES

ogc:contact includes "role"

[Is static True. Containment is Not Specifi

2-02 temporalProgramAffiliation : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specifi

7-14 internationalExchange : Boolean Private

[Is static True. Containment is Not Specifi

7-14 internationalReportingSchedule : Private

complex type: when are observations reported internationally? Should be expressed as a temporal coverage

[Is static True. Containment is Not Specifi

id : Private

[Is static True. Containment is Not Specifi

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public observation (Class) observation
Cardinality: [1]

Target: Public deployment (Class) deployment
Cardinality: [0..*]

Association (direction: Unspecified)

Source: Public facility (Class) facility
Cardinality: [1]

Target: Public observation (Class) observation
Cardinality: [1..*]

procedure

Class in package 'WMDR2core'

Extension point to document
- ObservingProcedure
- (Raw)DataProcessing
- ...

pro
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ASSOCIATIONS

ASSOCIATIONS
ObservingProcedure(s)